$\oint$ $\boxed{1}$ Atomic Physics

Angular soln

Spherical harmonics $Y_{l,m}(\theta,\phi) \propto \underbrace{\cdots}_{\sin\theta,\ \cos\theta \ldots} e^{im\phi}$

Radial soln

Eigenenergies $E_n = -hcR_\infty \frac{Z^2}{n^2}$

$$|\Psi(r,\theta,\phi)|^2 = R_{n,l}^2(r)\,|Y_{l,m}(\theta,\phi)|^2$$

Key constants

$$hcR_\infty = \frac{m_e}{2\hbar^2}\left(\frac{e^2}{4\pi\varepsilon_0}\right)^2 \simeq \underline{13.6\,eV}$$

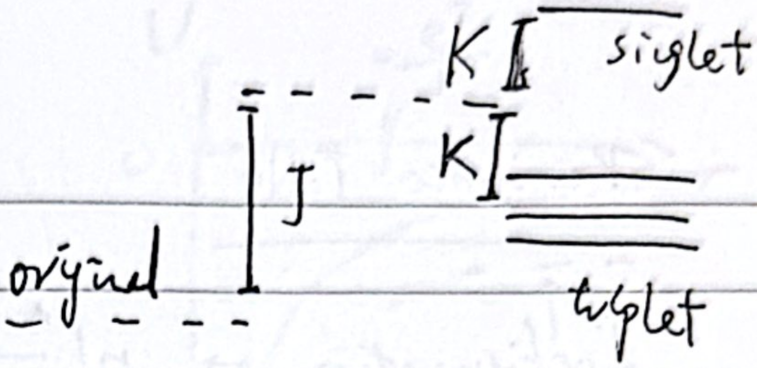
(FSC) $\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \simeq \frac{1}{137}$

(Bohr Magnetron) $\mu_B = \frac{e\hbar}{2m_e} \simeq 9.27\times10^{-24}\ JT^{-1}$

Transitions

Selection rules: (dipole transition)
$$\begin{cases} \Delta l = \pm 1 \\ \Delta m = \underset{\downarrow}{0},\ \underset{\downarrow}{\pm 1} \end{cases}$$

$\underline{\hat{z}}$ $\hat{x}, \underline{\hat{y}}$



Helium

He. Excited states

degenerate P.T. → 2 identical e^-

$$\Delta E = J \pm K$$

J: direct integral; K: exchange integral

→ $\Delta E_+ = J + K$ (singlet)

$\Delta E_- = J - K$ (triplet)

- $\psi_{spatial}\,\psi_{spin}$ antisymmetric → (SA or AS)

Quantum Defect (Alkalis)

- Degeneracy is broken for atoms of $> 1 e^-$. (same n)
- s-electrons penetrate deep, less 'screening'. (substantial wavefunction at core)

lower energy than expected. (ie. 4s lower than 3d)

$$E(n,l) = -hc\frac{R_\infty}{(\underbrace{n-\delta_l}_{n^* = n - \delta_l})^2}$$

δ_l: quantum defect

$n^* = n - \delta_l$

effective principal quantum number

Periodic Table

Configuration

Configuration $\rightarrow$ nl^{x} $\leftarrow$ # of e^-

principal; l: s, p, d, f ($l = 0, 1, 2, 3$) (C.F. approx.)

Term & Level

$$\underbrace{n\,{}^{(2S+1)}L}_{\text{term}}{}_J \leftarrow \text{Levels}$$

term (residue elec.) — Levels (spin-orbit effects)

C.F. Approx.

$$H_{CF} \simeq \sum_i \left(-\frac{\hbar^2}{2m}\nabla_i^2 + V_{CF}(r_i) \right)$$

N-eqns with form

$$\left(-\frac{\hbar^2}{2m}\nabla_i^2 + V_{CF}(r_i)\right)\psi_i = E_i\psi_i$$

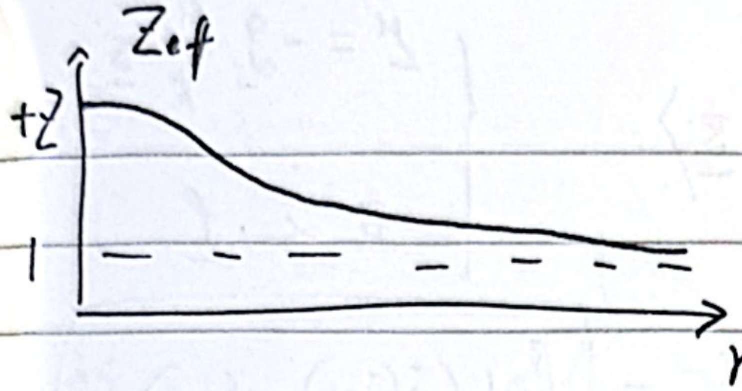
$$E_{atom} = E_1 + E_2 + \cdots + E_N$$

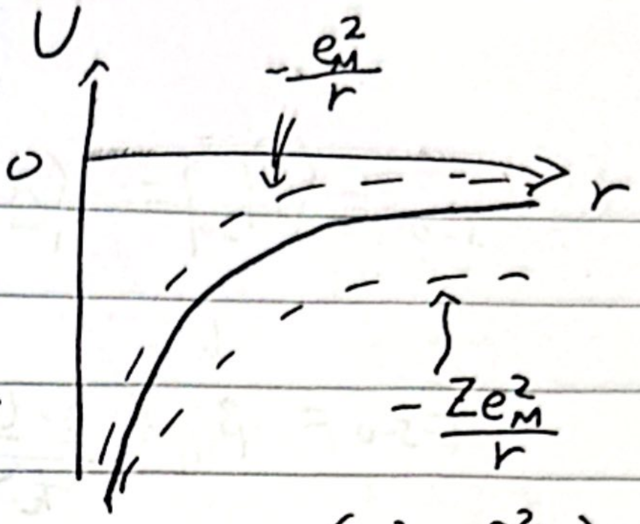
$$\psi_i = R(r_i)\underbrace{\psi_{l_i, m_i}\psi_{spin}(i)}_{\sim\ \text{hydrogenic}}$$

Qualitatively,

$r \to 0$, full charge seen, $\underline{E} \to \frac{Ze}{4\pi\epsilon_0 r^2}\hat{r}$

$r \to \infty$, well-screened, $\underline{E} \simeq \frac{e}{4\pi\epsilon_0 r^2}\hat{r}$





$\left(e_M^2 = \frac{e^2}{4\pi\varepsilon_0}\right)$

Perturbation Sequence

$$\hat{H}_{atom} = \hat{H}_{CF} + \hat{H}_{RE} + \hat{H}_{SO} + \hat{H}_{HFS} \rightarrow e^- \text{ \& nucleus interact (h.f.s.)}$$

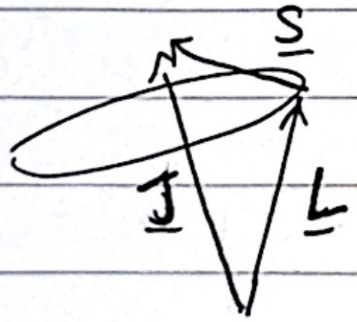
- $\hat{H}_{CF}$ ↓ Central Field (nl) — configuration
- $\hat{H}_{RE}$ ↓ Residue Electrostatic $^{(2S+1)}L$ — term
- $\hat{H}_{SO}$ ↓ Spin-orbit Coupling $^{(2S+1)}L_J$ — level

$(L = \sum_i l_i,\ S = \sum_i s_i)$

Spin-Orbit

(Alkalis) $\longrightarrow$ $\hat{H}_{RE}$ incorporated into C.F. Approx.

Vector model



$$\underline{J} = \underline{L} + \underline{S} \qquad \boxed{|L-S| \leq J \leq L+S}$$

↓

$$\underline{J}^2 - \underline{L}^2 - \underline{S}^2 = 2\,\underline{S}\cdot\underline{L}$$

$$\langle \underline{S}\cdot\underline{L} \rangle = \frac{\hbar^2}{2}\left(J(J+1) - L(L+1) - S(S+1)\right)$$

$$E_{s-o} = \langle \hat{H}_{so} \rangle = -\langle \underline{\mu} \cdot \underline{B} \rangle \qquad \left\{ \begin{array}{l} \underline{\mu} = -g_s \frac{\mu_B}{\hbar} \underline{s} \\ \underline{B} \propto \underline{l} \end{array} \right.$$

$$\downarrow$$

$$E_{s-o} = \beta_{nl} \frac{\langle \underline{s} \cdot \underline{l} \rangle}{\hbar^2} = \boxed{\frac{\beta_{nl}}{2}\left(j(j+1) - s(s+1) - l(l+1) \right)}$$

where $\beta_{nl} = \frac{\mu_0 Z^4 g_s \mu_B^2}{4\pi} \frac{1}{n^3 a_0^3 l(l+\frac{1}{2})(l+1)}$

Interval Rule

$$\Delta E_{j,j-1} = \frac{\beta_{nl}}{2}\left(j(j+1) - (j-1)j \right) = \underline{\underline{\beta_{nl} j}}$$

L–S Coupling Scheme

$$\underline{L} = \Sigma_i \underline{l_i} \qquad \underline{S} = \Sigma_i \underline{s_i} \qquad \underline{J} = \underline{L} + \underline{S}$$

where C.F. & R.E. dominates over S-O

L, S, J, m_J good quantum #

(m_s, m_L NOT!)

Magnetic Fields (External)

$$\hat{H}_{mag.} = -\underline{\mu}_{atom} \cdot \underline{B}_{ext}$$

$$\left\{ \begin{array}{lll} \text{① weak } B & & \hat{H}_{mag} < \hat{H}_{s-o} \\ \text{② strong } B & & \hat{H}_{mag} > \hat{H}_{s-o} \end{array} \right.$$

① weak field, no spin (Normal Zeeman)

$|n, L, S, J, M_J\rangle$ $\qquad S=0$

$S=0,\ J=L \rightarrow\ L\ \&\ M_L$ sufficient

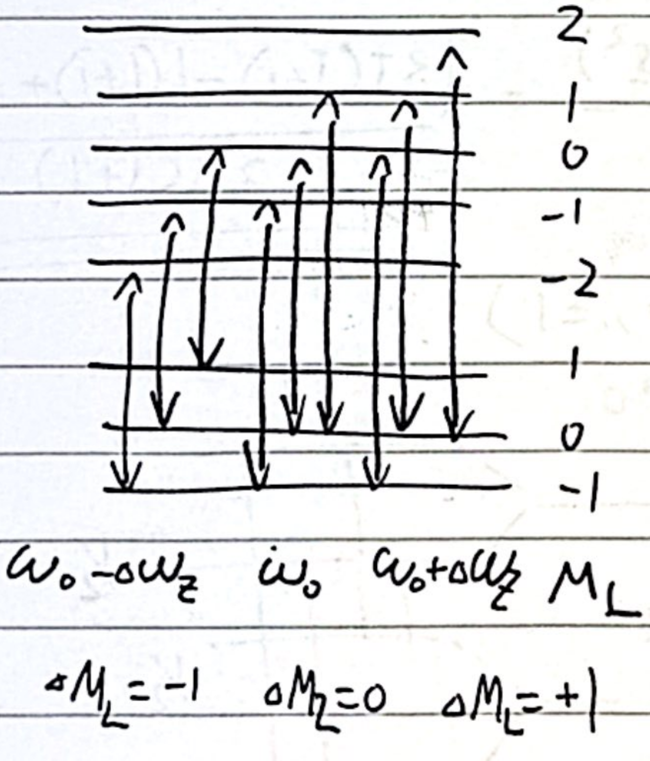
$(-L \leq M_L \leq L)$

$$\underline{\mu}_L = -g_L \mu_B \underline{L}$$

$$\Delta E_Z = -\langle \underline{\mu}_L \cdot \underline{B}_{ext} \rangle = g_L \mu_B \langle \underline{L} \cdot \underline{B} \rangle = g_L \mu_B B M_L$$

Selection rules $\Delta M_L = 0, \pm 1$ only.

$\times\ 0 \leftrightarrow 0$



$$\Delta\omega_Z = \frac{\Delta E_Z'}{\hbar} = \frac{\mu_B B}{\hbar}$$

Lorentz triplet

② weak field, spin-orbit ✓ (Anomalous Zeeman)

$|n, L, S, J, M_J\rangle$ $\qquad \underline{J} = \underline{L} + \underline{S} \qquad S \neq 0$

$$\langle \hat{H} \rangle = \Delta E_{AZ} = g_L \mu_B \langle \underline{L} \cdot \underline{B} \rangle + g_S \mu_B \langle \underline{S} \cdot \underline{B} \rangle$$

$$= g_J \mu_B \langle \underline{J} \cdot \underline{B} \rangle = g_J \mu_B (M_J B)$$

Calculate Landé g-factor g_J

$$g_J M_J = g_L L_J + g_S S_J$$

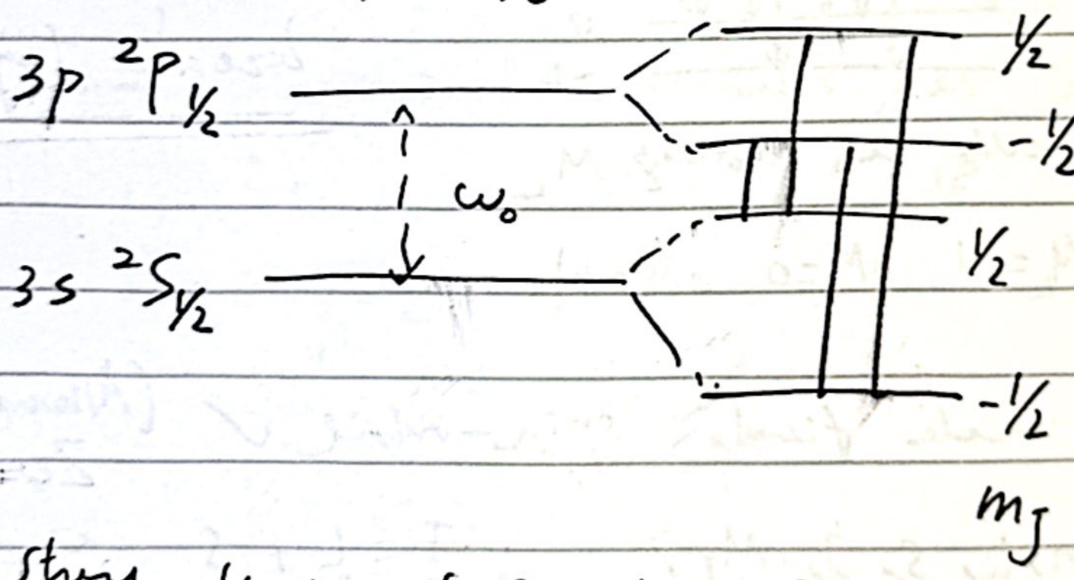
$$= g_L \frac{|\underline{L}\cdot\underline{J}|}{|\underline{J}|^2} M_J + g_S \frac{|\underline{S}\cdot\underline{J}|}{|\underline{J}|^2} M_J$$

$$g_J = \underbrace{g_L}_{1} \frac{\frac{1}{2}(\underline{L}^2+\underline{J}^2-\underline{S}^2)}{|\underline{J}|^2} + \underbrace{g_S}_{2} \frac{\frac{1}{2}(\underline{S}^2+\underline{J}^2-\underline{L}^2)}{|\underline{J}|^2}$$

$$= \frac{(3\underline{J}^2-\underline{L}^2+\underline{S}^2)}{2|\underline{J}|^2} = \boxed{\frac{3J(J+1)-L(L+1)+S(S+1)}{2J(J+1)}}$$

(still $\Delta M_J = 0, \pm 1$)

$\times\ 0 \leftrightarrow 0$



③ strong field, S-O ✓ (Paschen-Back)

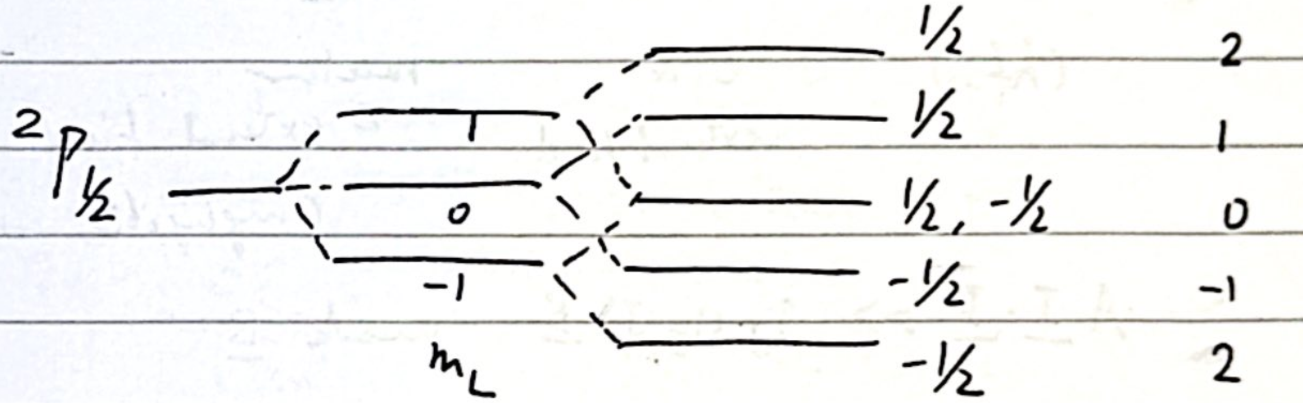
$\underline{\mu}_L$ & $\underline{\mu}_S$ precess rapidly about $\underline{B}_{ext}$ more so than about each other.

- $\underline{J}$ undefined
- Use $\{L, M_L\}$, $\{S, M_S\}$ good quantum #s.

$$\hat{H} = g_L \mu_B \underline{L}\cdot\underline{B} + g_s \mu_B \underline{S}\cdot\underline{B}$$

$$\Delta E_{PB} = (M_L + 2M_S)\mu_B B$$

- split into $(2L+1)$ levels (M_L label) then split into 2 levels for each M_L. (M_S label)



selection rules:

$\Delta M_L = 0, \pm 1$, $\Delta S = 0$

$\times\ 0 \leftrightarrow 0$ $\quad (\Delta M_S = 0)$

HFS

$$\hat{H}_{hfs} = A\,\underline{I}\cdot\underline{J} \qquad \underline{I} + \underline{J} = \underline{F} \qquad |\underline{I}-\underline{J}| \leq |\underline{F}| \leq |\underline{I}+\underline{J}|$$

$$\Delta E = \frac{A}{2}\left(\underline{F}^2 - \underline{J}^2 - \underline{I}^2\right) = \frac{A}{2}\left(F(F+1) - I(I+1) - J(J+1)\right)$$

good quantum # $\rightarrow$ $|I, J, F, M_F\rangle$

Selection rule $\Delta F = 0, \pm 1 \quad \times\ 0 \leftrightarrow 0$

Interval rule

$$\Delta E_F = AF$$

\# of spectral lines $\begin{cases} (2I+1) & I<J \\ (2J+1) & I>J \end{cases}$

HFS & Magnetic Fields

$$\underset{\text{(hfs)}}{A_J \underline{I}\cdot\underline{J}} + \underset{\substack{e^- \,\& \\ \text{external field}}}{g_J \mu_B \underline{J}\cdot\underline{B}} - \underset{\substack{\text{nuclear} \\ \text{\& external field} \\ \text{(negligible, } \to 0)}}{g_I \mu_N \underline{I}\cdot\underline{B}}$$

$A\,\underline{I}\cdot\underline{J} \gg g_J \mu_B \underline{J}\cdot\underline{B}$ weak $\underline{B}$

$A\,\underline{I}\cdot\underline{J} \ll g_J \mu_B \underline{J}\cdot\underline{B}$ strong $\underline{B}$

① weak field

$|n, L, s, J, I, F, M_F\rangle \qquad \underline{I}+\underline{J}=\underline{F}$

$$\Delta E_F = \langle \underset{\substack{\downarrow \\ -g_F\mu_B \underline{F}}}{\underline{\mu}_F} \cdot \underline{B}\rangle = g_F \mu_B \langle \underline{F}\cdot\underline{B}\rangle = g_J \mu_B \frac{\langle \underline{J}\cdot\underline{F}\rangle}{|\underline{F}|^2}\langle \underline{F}\cdot\underline{B}\rangle$$

$$\underline{J}\cdot\underline{F} = \frac{1}{2}(\underline{F}^2+\underline{J}^2-\underline{I}^2)$$

nuclear factor: $g_F = g_J\left\{\dfrac{F(F+1)+J(J+1)-I(I+1)}{2F(F+1)}\right\}$

$\Delta E_F = g_F \mu_B B_{ext} M_F$

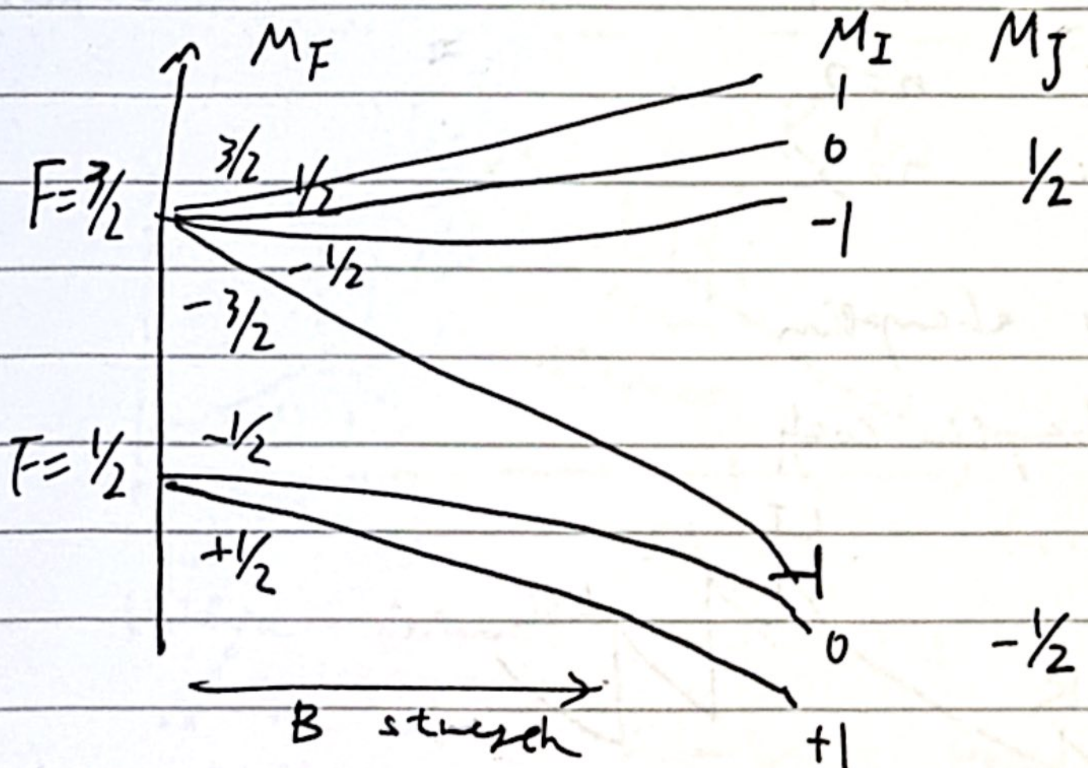
② strong field (Back-Goudsmit effect)

$\underline{I}$ & $\underline{J}$ precess rapidly with B_{ext}

$\underline{F}$ undefined (so as M_F)

$\Delta E_{BG} = g_J \mu_B M_J B + A M_I M_J$

- each M_J splits into $2I+1$ levels again.



i.e. $\underline{I=1,\ J=1/2}$ case

(g_F negative for lower substates)

X-rays (1 ~ 100 keV roughly)

Transition

$$\Delta E_{ij} = hcR\left(\frac{(Z-\sigma_i)^2}{n_i^2} - \frac{(Z-\sigma_j)^2}{n_j^2}\right)$$

- σ considers both internal & external screening

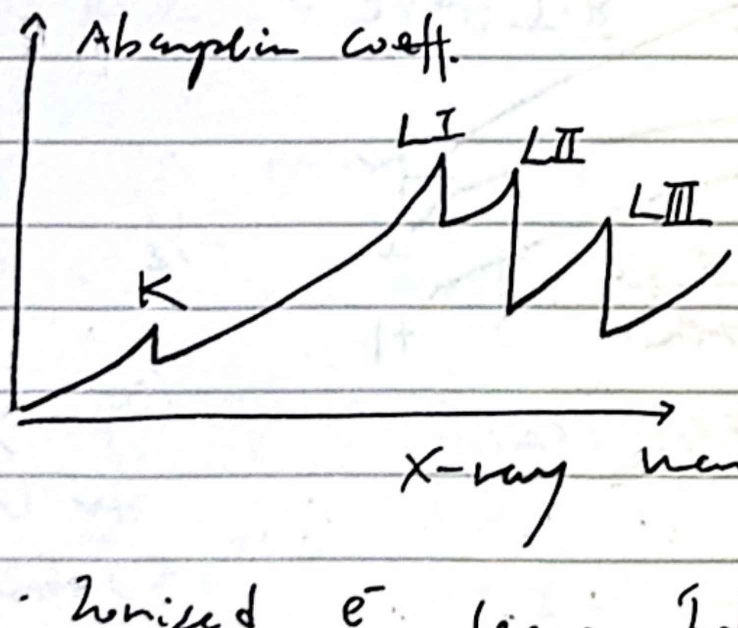
(whether e^- passes around other electron clouds)

K : $n=1$

L : $n=2$

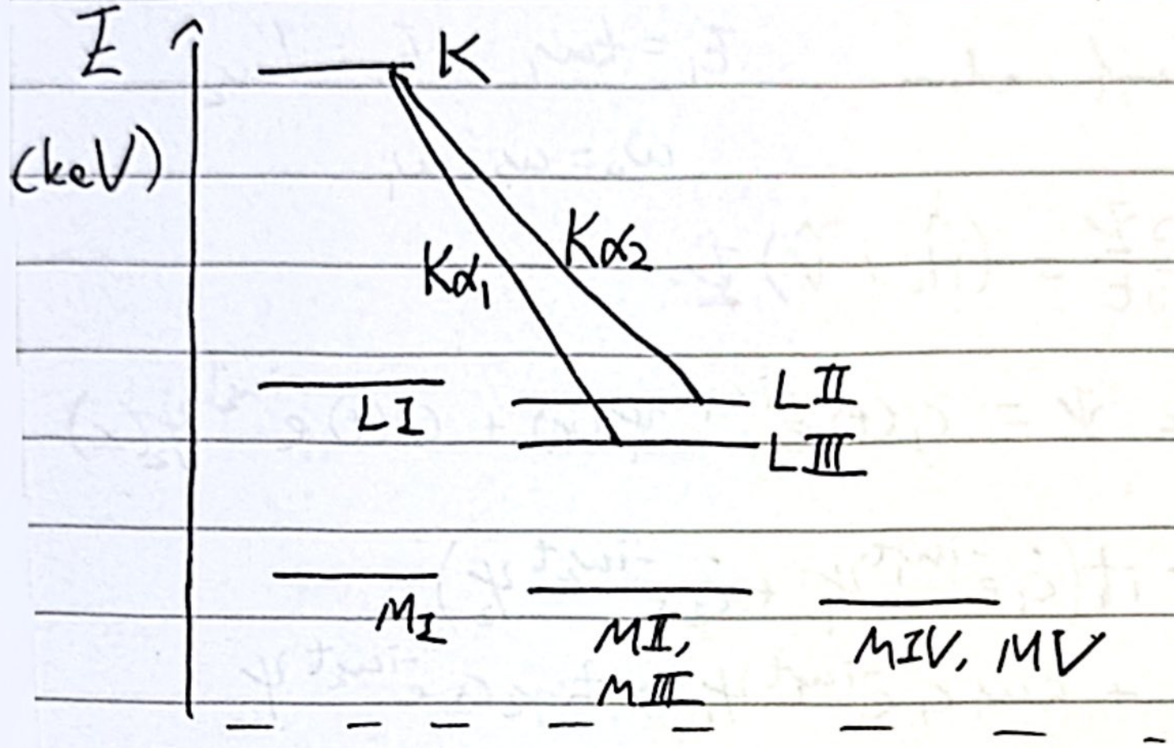
M : $n=3$

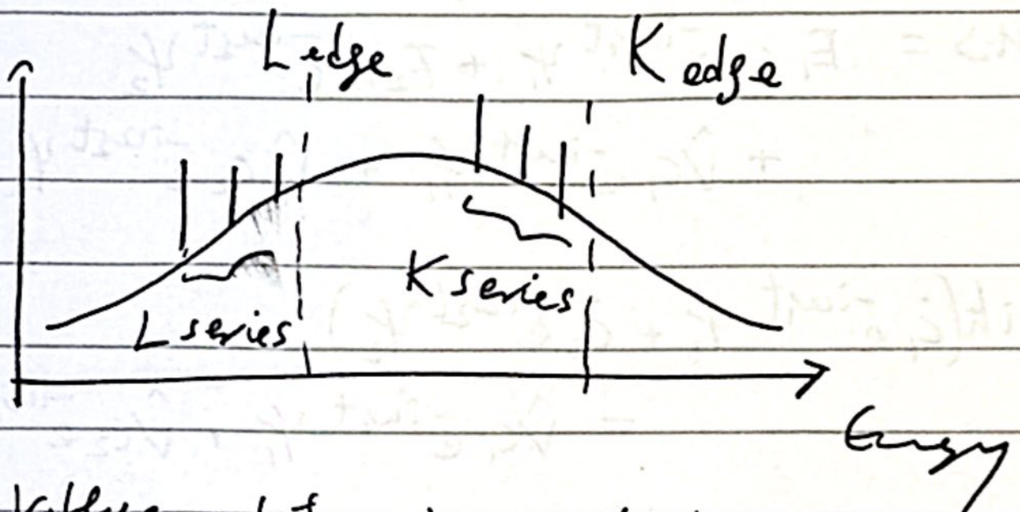
X-ray absorption



- Ionized e^- leave 'holes' in an inner shell.
- Higher-states e^- drop to fill holes &

leave new holes to be filled. → Several Series of transitions.





- Voltage determines which e^- is emitted (characteristic radiation)

- Continuum corresponds to e^- decelerated through material & radiating.

Auger Effect

An e^- drops to a lower hole with remnant energy to excite another e^- too.

Light-Matter Interactions

two-level atom $\quad E_1 = \hbar\omega_1 \quad E_2 = \hbar\omega_2$

$$\omega_0 = \omega_2 - \omega_1$$

$$i\hbar\frac{\partial\Psi}{\partial t} = (\hat{H}_0 + \hat{V})\Psi$$

where $\Psi = c_1(t)e^{-i\omega_1 t}\psi_1(x) + c_2(t)e^{-i\omega_2 t}\psi_2(x)$

$$LHS = i\hbar\left(\dot{c}_1 e^{-i\omega_1 t}\psi_1 + \dot{c}_2 e^{-i\omega_2 t}\psi_2\right)$$
$$+ \hbar\omega_1 c_1 e^{-i\omega_1 t}\psi_1 + \hbar\omega_2 c_2 e^{-i\omega_2 t}\psi_2$$

$$RHS = E_1 c_1 e^{-i\omega_1 t}\psi_1 + E_2 c_2 e^{-i\omega_2 t}\psi_2$$
$$+ \hat{V}c_1 e^{-i\omega_1 t}\psi_1 + \hat{V}c_2 e^{-i\omega_2 t}\psi_2$$

$$i\hbar\left(\dot{c}_1 e^{-i\omega_1 t}\psi_1 + \dot{c}_2 e^{-i\omega_2 t}\psi_2\right)$$
$$= \hat{V}c_1 e^{-i\omega_1 t}\psi_1 + \hat{V}c_2 e^{-i\omega_2 t}\psi_2$$

multiply by ψ_1^* & ψ_2^* & integrate over all space.

$\left(V_{nm} = \langle\psi_n|\hat{V}|\psi_m\rangle = \int_V \psi_n^* \hat{V}\psi_m \, d^3\underline{r}\right)$

$$\begin{cases} \dot{c}_1 = -\frac{i}{\hbar}\left(c_1 V_{11} + c_2 V_{12} e^{-i\omega_0 t}\right) \\ \dot{c}_2 = -\frac{i}{\hbar}\left(c_1 V_{21} e^{+i\omega_0 t} + c_2 V_{22}\right) \end{cases}$$

· rate eqns

Consider $\mathcal{E} = \mathcal{E}_0 \cos(\omega t)$

$$\hat{V}(t) = -\underline{p}\cdot\underline{\mathcal{E}} = -(-e\underline{r}\cdot\underline{\mathcal{E}}) = ex\mathcal{E}_0\cos(\omega t)$$

$V_{11} = V_{22} = 0$ (odd-parity operator x)

<u>def</u> $\Omega = \frac{\mathcal{E}_0}{\hbar}\langle\psi_1|ex|\psi_2\rangle$ (Rabi frq.)

$$V_{12}^* = V_{12} = V_{21} = \langle\psi_1|ex|\psi_2\rangle\mathcal{E}_0\cos(\omega t)$$
$$= \hbar\Omega\cos(\omega t)$$

$$\begin{cases} \dot{c}_1 = -i\Omega\cos(\omega t)\,e^{-i\omega_0 t}\,c_2 \\ \dot{c}_2 = -i\Omega\cos(\omega t)\,e^{+i\omega_0 t}\,c_1 \end{cases} \quad (\star)$$

1 <u>weak field</u>

low transition prob.

$c_2(t) \ll c_1(t) \qquad c_1(t) \simeq 1$

$$\dot{c}_2 \simeq -i\Omega\cos(\omega t)\,e^{i\omega_0 t}$$
$$\simeq -\frac{i\Omega}{2}\left(e^{i(\omega+\omega_0)t} + e^{-i(\omega-\omega_0)t}\right)$$

RWA $\rightarrow |\omega-\omega_0| \ll (\omega+\omega_0)$

$$\dot{c}_2 \simeq -\frac{i\Omega}{2}e^{-i(\omega-\omega_0)t}$$

integrate to get

$$c_2(t) = -\frac{1}{2}\Omega\left(\frac{1-e^{-i(\omega-\omega_0)t}}{\omega-\omega_0}\right)$$

$$= -\frac{1}{2}\Omega e^{-i(\omega-\omega_0)t/2}\left(\frac{e^{i(\omega-\omega_0)t/2}-e^{-i(\omega-\omega_0)t/2}}{(\omega-\omega_0)}\right)$$

$$|c_2(t)|^2 = \left(\frac{\Omega}{2}\right)^2\left(\frac{\sin(\frac{1}{2}(\omega-\omega_0)t)}{\frac{1}{2}(\omega-\omega_0)}\right)^2$$

Consider $\rho(\omega)d\omega = \frac{1}{2}\varepsilon_0 \mathcal{E}_0^2$

$$|c_2(t)|^2 = \frac{|\langle\psi_1|ex|\psi_2\rangle|^2}{2\varepsilon_0\hbar^2}\int_{\omega_0-\Delta\omega}^{\omega_0+\Delta\omega}\rho(\omega)\left(\frac{\sin(\frac{1}{2}(\omega-\omega_0)t)}{\frac{1}{2}(\omega-\omega_0)}\right)^2 d\omega$$

$\rho(\omega)=\rho(\omega_0)$ mostly,

so extend int. to $[-\infty,\infty)$ is ok.

$$\pi t \simeq 2\pi t$$

$$|c_2(t)|^2 = \frac{\pi t}{\varepsilon_0\hbar^2}|\langle\psi_1|ex|\psi_2\rangle|^2\rho(\omega_0)$$

Unpolarized has

$$R_{12} = \frac{1}{3}\frac{|c_2(t)|^2}{t} = \underbrace{\frac{\pi}{3\varepsilon_0\hbar^2}|\langle\psi_1|ex|\psi_2\rangle|^2}_{B_{12}}\rho(\omega_0)$$

$$A_{21} = \underbrace{\frac{\hbar\omega_0^3}{\pi^2 c^3}}_{\rho_0} B_{12} = \frac{\omega_0^3}{3\pi\varepsilon_0 \hbar c^3} |\langle\psi_1|ex|\psi_2\rangle|^2$$

$\boxed{2}$ Strong field limit

$$\circledast \rightarrow \begin{cases} \dot{c}_1 = -\frac{1}{2} i\Omega e^{i\delta t} c_2 \\ \dot{c}_2 = -\frac{1}{2} i\Omega e^{-i\delta t} c_1 \end{cases}$$

RWA, $\delta = \omega - \omega_0$ (detuning) ...

- $\delta = 0$, resonant excitation

$$\ddot{c}_2 = -\frac{1}{4}\Omega^2 c_2 \quad [c_2(0) = 0 \text{ initially}]$$

$$c_2(t) = \sin\left(\frac{\Omega}{2} t\right)$$

$$\downarrow$$

$$c_1(t) = i\cos\left(\frac{\Omega}{2} t\right)$$

$$\begin{cases} |c_1(t)|^2 = \cos^2\left(\frac{1}{2}\Omega t\right) \\ |c_2(t)|^2 = \sin^2\left(\frac{1}{2}\Omega t\right) \end{cases}$$

period $\frac{2\pi}{\Omega}$

Rabi flopping

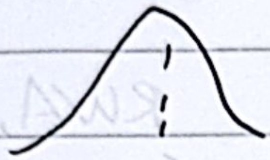
- $\delta \neq 0$, solve the rate eqns.

$$\rightarrow |c_2(t)|^2 = \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2\left(\frac{t}{2}\sqrt{\Omega^2 + \delta^2}\right)$$

more rapid angular freq $\rightarrow \sqrt{\Omega^2 + \delta^2}$.

§12 Laser Physics

Lineshapes

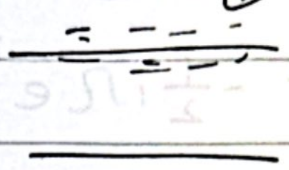


$g(\omega)$

↑

normalised lineshape.

blurred



e

g

Homogeneous Broadening

· Affects all atoms in the same way.

① (Natural) Lifetime Broadening (light emitted by oscillating dipoles)

$$P = -e\langle\psi|z|\psi\rangle$$

$$|\psi\rangle = \alpha_g|g\rangle + \alpha_e|e\rangle$$

$$\langle z\rangle = \underbrace{\alpha_g^*\langle g|z|e\rangle}_{\sim e^{-i\omega_0 t}} + \underbrace{\alpha_g\alpha_e^*\langle e|z|g\rangle}_{\sim e^{i\omega_0 t}} \propto \cos(\omega_0 t)$$

Assume $P_e \propto e^{-t/\tau} = e^{-\gamma t}$ (exp. decay)

$$\alpha_e \propto e^{-\gamma/2\, t}$$

So $\langle z\rangle \propto \cos(\omega_0 t)\, e^{-\frac{\gamma}{2}t}$.

F. T. (keep Re.)

$$\int_0^\infty e^{-\frac{\gamma t}{2}} e^{-i\omega t}\, dt = \frac{1}{\gamma/2 + i\omega} = \frac{\gamma/2 - i\omega}{\left(\frac{\gamma}{2}\right)^2 + \omega^2}$$

$$g(\omega) = \frac{1}{\pi} \frac{\gamma/2}{(\gamma/2)^2 + \omega^2}$$

HWHM = $\gamma/2$

FWHM = $\gamma = \frac{1}{\tau}$

$\Delta\omega_N$

Natural linewidth of transition $k \to j$:

$$\Delta\omega_N = \gamma = \frac{1}{\tau_k} + \frac{1}{\tau_j}$$

② Pressure Broadening (due to collisions with other atoms, ions, etc.)

$$\alpha_e \propto e^{-\left(\frac{\gamma}{2} + \frac{1}{\tau_c}\right)t}$$

$\downarrow$ F.T.

$g(\omega)$ also Lorentzian

with $\Delta\omega_p = \gamma + \frac{2}{\tau_c}$

③ Phonon Broadening

- Energy level fluctuation due to thermal motion & dependent on T & material.

Inhomogeneous Broadening

① Doppler Broadening

freq. shift $= \omega - \omega_0 = \frac{v_z}{c}\omega_0$

Maxwellian dist. for velocities:

$$P(v_z)\,dv_z \propto \exp\left(-\frac{Mv_z^2}{2k_BT}\right)dv_z$$

$$\propto \exp\left(-\frac{Mc^2}{2k_BT}\left(\frac{\omega-\omega_0}{\omega_0}\right)^2\right)dv_z$$

find FWHM → set $\exp(\ldots) = \frac{1}{2}$

$$\frac{Mc^2}{2k_BT}\left(\frac{\omega-\omega_0}{\omega_0}\right)^2 = \ln 2$$

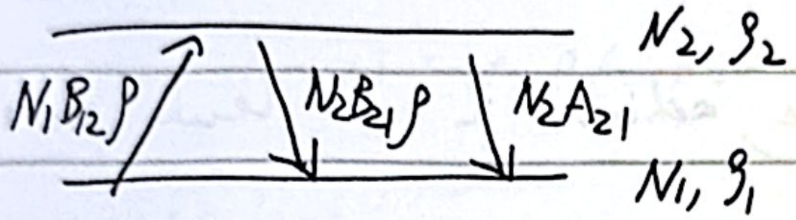
$$(\underbrace{\omega-\omega_0}_{\frac{\Delta\omega_D}{2}})^2 = 2\ln 2\,\frac{\omega_0^2}{c^2}\frac{k_BT}{M}$$

$$\boxed{\Delta\omega_D = 2\sqrt{2\ln 2}\,\frac{\omega_0}{c}\sqrt{\frac{k_BT}{M}}}$$

② Amorphous solids

- Active ions in different locations of a non-uniform solid experience different local crystal fields & other environmental factors → different broadening response.

Pop Inversion



2 Levels model

$$\dot{N}_2 = N_1 B_{12}\rho - N_2 B_{21}\rho - N_2 A_{21}$$

$$\dot{N}_1 = -\dot{N}_2 \quad (= 0) \text{ steady state}$$

$$N_1 B_{12}\rho = N_2 B_{21}\rho + N_2 A_{21}$$

$$\frac{N_2}{N_1} = \frac{B_{12}\rho}{B_{21}\rho + A_{21}} \xrightarrow[\text{high } \rho \text{ (intensity)}]{} \frac{B_{12}}{B_{21}}$$

Thermal equilibrium assumes

$$\frac{N_2}{N_1} = \frac{g_2}{g_1} e^{-\frac{\hbar\omega}{k_B T}} \quad \& \quad \rho = \frac{\rho_0}{e^{\hbar\omega/k_B T} - 1} \quad \left(\rho_0 = \frac{\hbar\omega^3}{\pi^2 c^3}\right)$$

substitute to get

$$B_{12} = \left(B_{21} + \frac{A_{21}}{\rho_0}\left(e^{\frac{\hbar\omega}{k_B T}} - 1\right)\right)\frac{g_2}{g_1} e^{-\frac{\hbar\omega}{k_B T}}$$

$$= \frac{g_2}{g_1}\frac{A_{21}}{\rho_0} + \frac{g_2}{g_1}\left(B_{21} - \frac{A_{21}}{\rho_0}\right) e^{-\frac{\hbar\omega}{k_B T}}$$

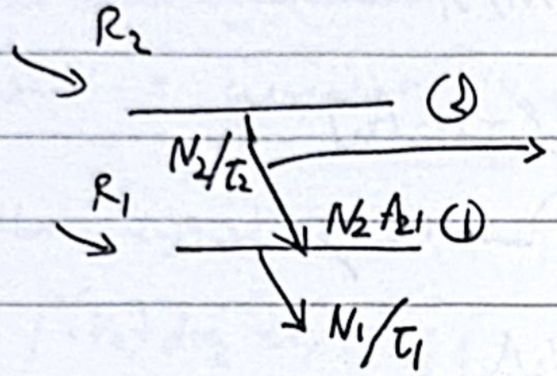
① $\underline{B_{21} = \frac{A_{21}}{\rho_0}}$ (for all T)

② $\underline{B_{12} = \frac{g_2}{g_1} B_{21}}$

- true for all cases, not just thermal equil.

Pumping

· Inv. cannot be achieved in 2 levels alone.



$$\begin{cases} \dot{N}_2 = R_2 - \dfrac{N_2}{\tau_2} \\ \dot{N}_1 = R_1 + N_2 A_{21} - \dfrac{N_1}{\tau_1} \end{cases}$$

$$\dot{N}_1 = \dot{N}_2 = 0$$

$$\begin{cases} N_2 = R_2 \tau_2 \\ N_1 = R_1 \tau_1 + N_2 A_{21} \tau_1 = R_1 \tau_1 + R_2 \tau_2 A_{21} \tau_1 \end{cases}$$

Need $N_2 B_{21} \rho > N_1 B_{12} \rho$

$$N_2 B_{21} > N_1 \frac{g_2}{g_1} B_{21}$$

$$\cancel{B} \quad \underline{\underline{\frac{N_2}{N_1} > \frac{g_2}{g_1}}} \quad \text{or} \quad \underline{\frac{N_2}{g_2} > \frac{N_1}{g_1}}$$

$$\frac{R_2 \tau_2}{g_2} > \frac{R_1 \tau_1}{g_1} + \frac{R_2 \tau_2 A_{21} \tau_1}{g_1}$$

$$\underbrace{\frac{R_2}{R_1} \frac{\tau_2}{\tau_1} \frac{g_1}{g_2}}_{\text{high, +ve}} \underbrace{\left(1 - \frac{g_2}{g_1} A_{21} \tau_1\right)}_{\text{+ve}} > 1$$

<u>Necessary condition</u> $\rightarrow$ $\underline{A_{21} < \dfrac{g_1}{g_2} \dfrac{1}{\tau_1}}$

Gain

$$I(\omega, z) = c\,\rho(\omega, z)$$

I: spectral irradiance/intensity $Wm^{-2}Hz^{-1}$

ρ: spectral energy density $Jm^{-3}Hz^{-1}$

$$\text{Rate} = (N_2B_{21} - N_1B_{12})\,g_H(\omega-\omega_0)\,\rho(\omega,z)\,d\omega\,A\,dz$$

$$\text{Power} = \delta I\,d\omega\,A$$

$$= \hbar\omega(N_2B_{21} - N_1B_{12})\,g_H(\omega-\omega_0)\underbrace{\rho(\omega,z)}_{I(\omega,z)/c}\,d\omega\,A\,dz$$

so
$$\frac{\partial I}{\partial z} = \frac{\hbar\omega}{c}(N_2B_{21} - N_1B_{12})\,g_H(\omega-\omega_0)\,I(\omega,z)$$

$$= \underbrace{\left(N_2 - \frac{g_2}{g_1}N_1\right)}_{N^*}\underbrace{B_{21}\,g_H(\omega-\omega_0)\frac{\hbar\omega}{c}}_{\sigma_{21}(\omega-\omega_0)}\,I$$

N^*: pop. inversion density

$\sigma_{21}(\omega-\omega_0)$: optical gain coefficient

$$= \underbrace{N^*\,\sigma_{21}(\omega-\omega_0)}_{\alpha_{21}(\omega-\omega_0)}\,I$$

$\alpha_{21}(\omega-\omega_0)$: small gain coefficient

$$\left(\sigma_{21}(\omega-\omega_0) = \frac{\hbar\omega}{c}B_{21}\,g_H(\omega-\omega_0) = \frac{\pi^2c^2}{\omega^2}A_{21}\,g_H(\omega-\omega_0)\right)$$

Saturation

Now at steady state,

$$\begin{cases} \dfrac{dN_1}{dt} = R_1 + \underbrace{N^* \sigma_{21} \dfrac{I}{\hbar\omega}} + N_2 A_{21} - \dfrac{N_1}{\tau_1} = 0 \\[2ex] \dfrac{dN_2}{dt} = R_2 - \underbrace{N^* \sigma_{21} \dfrac{I}{\hbar\omega}}_{\star} - \dfrac{N_2}{\tau_2} = 0 \end{cases}$$

$$N^* = N_2 - \frac{g_2}{g_1} N_1$$

$$= \left(R_2\tau_2 - N^* \sigma_{21} \frac{I}{\hbar\omega} \tau_2\right)$$

$$(R_2\tau_2 - N^* \sigma_{21} \frac{I}{\hbar\omega} \tau_2)$$

$$- \frac{g_2}{g_1}\left(R_1\tau_1 + N^* \sigma_{21} \frac{I}{\hbar\omega}\tau_1 + N_2 A_{21} \tau_1\right)$$

$\downarrow$ ...

$$N^* = \frac{R_2\tau_2\left(1 - \frac{g_2}{g_1} A_{21}\tau_1\right) - \frac{g_2}{g_1} R_1 \tau_1}{1 + \sigma_{21}\frac{I}{\hbar\omega}\left(\tau_2 + \frac{g_2}{g_1}\tau_1 - \frac{g_2}{g_1} A_{21}\tau_1\tau_2\right)}$$

$$N^* = \frac{N^*(0)}{1 + I/I_s}$$

$$I_s = \frac{\hbar\omega}{\sigma_{21}\underbrace{\left(\tau_2 + \frac{g_2}{g_1}\tau_1 - \frac{g_2}{g_1}A_{21}\tau_1\tau_2\right)}_{\tau_R}} = \frac{\hbar\omega}{\sigma_{21}\tau_R}$$

saturated intensity
(at which N^* is halved)

recovery time

$$\tau_R \simeq \tau_2 \quad \text{if either} \begin{cases} \tau_1 << \tau_2 \\ A_{21} \simeq \dfrac{1}{\tau_2} \end{cases}$$

$$I_s \simeq \frac{\hbar\omega}{\sigma_{21}\tau_2}$$

Saturated gain

$$\alpha_I = N_I^* \sigma_{21} = \frac{N^*(0)\,\sigma_{21}}{1 + I/I_s} = \frac{\alpha(0)}{1 + I/I_s}$$

not (small signal gain)

Beam growth

$$\frac{dI}{dz} = \alpha_I I = \frac{\alpha_0}{1 + I/I_s} I$$

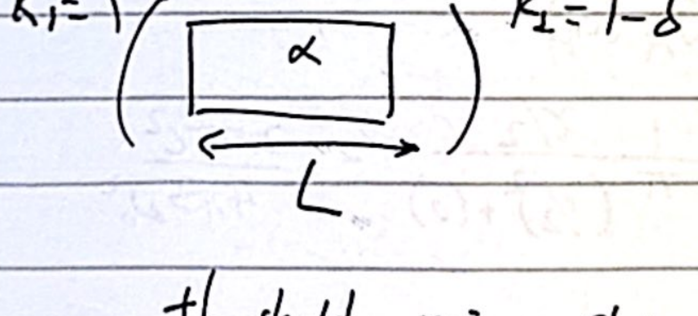
$$\downarrow$$

$$\left(\frac{1}{I} + \frac{1}{I_s}\right) dI = \alpha_0 dz$$

$$\ln\left(\frac{I(z)}{I(0)}\right) + \frac{I(z) - I(0)}{I_s} = \alpha_0 z$$

Analytic exact soln.

Threshold Effects in Cavity



$$e^{2\alpha L} R_1 R_2 > 1$$

for beam growth

threshold min α_{th} $\rightarrow$

$$2\alpha_{th} L \simeq -\ln(R_2) \simeq -\ln(1-\delta) \simeq \delta$$

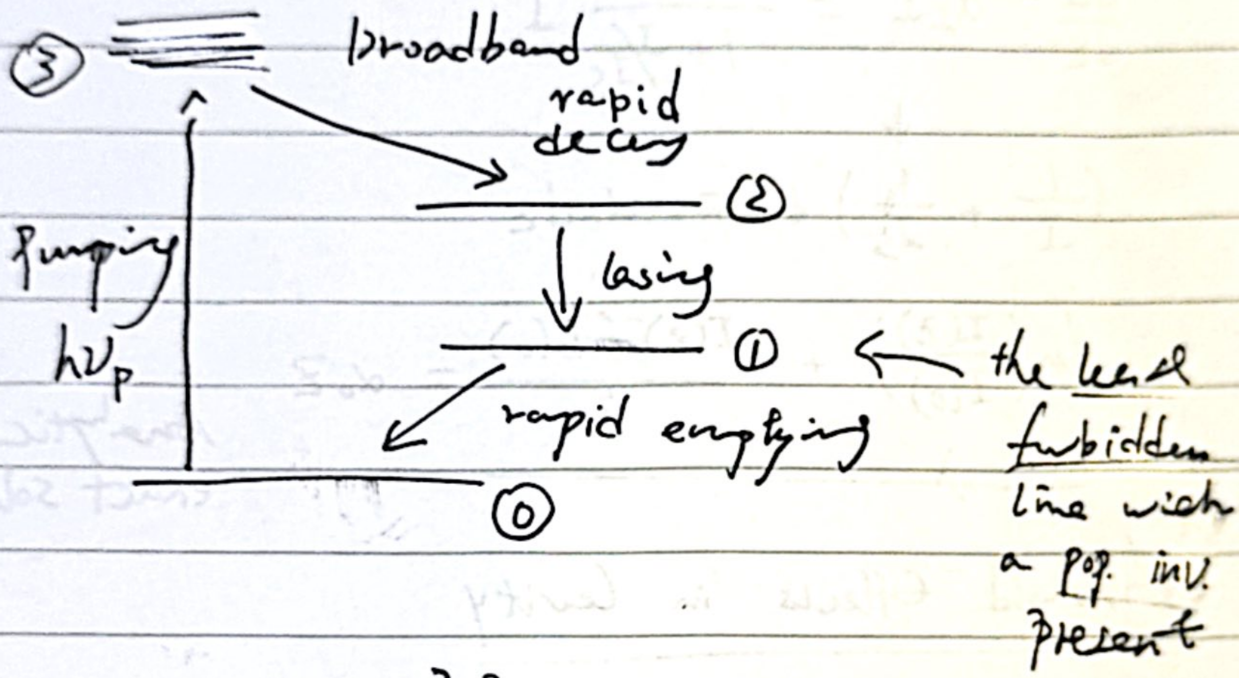
$$\boxed{\alpha_{th} \simeq \frac{\delta}{2L}}$$

$$\alpha_I \simeq \alpha_{th} \simeq \frac{\alpha_0}{1+I/I_s} \simeq \frac{\delta}{2L}$$

$$I = I_s\left(\frac{2\alpha_0 L}{\delta} - 1\right)$$

Multi-level Laser Designs

4-Level Laser (Nd:YAG)



$$\sigma_{21}(0) = \frac{\pi^2 c^2}{\omega_0^2} A_{21}\, g_H(0)$$

$$\simeq \frac{\pi^2 c^2}{\omega_0^2}\,\gamma\,\frac{1}{\pi}\,\frac{\gamma/2}{(\gamma/2)^2+(0)} = \frac{2\pi^2 c^2}{4\pi^3 \nu_0^2}$$

$$= \boxed{\frac{\lambda_0^2}{2\pi}}$$

Threshold energy (Nd:YAG)

$$R_1 R_2 \exp(\sigma N_{th}^* 2L) = 1$$ ☆

$N_1 \simeq 0$ $\quad N_2^{th} = \boxed{N_{th}^*}$

(no p.p.)

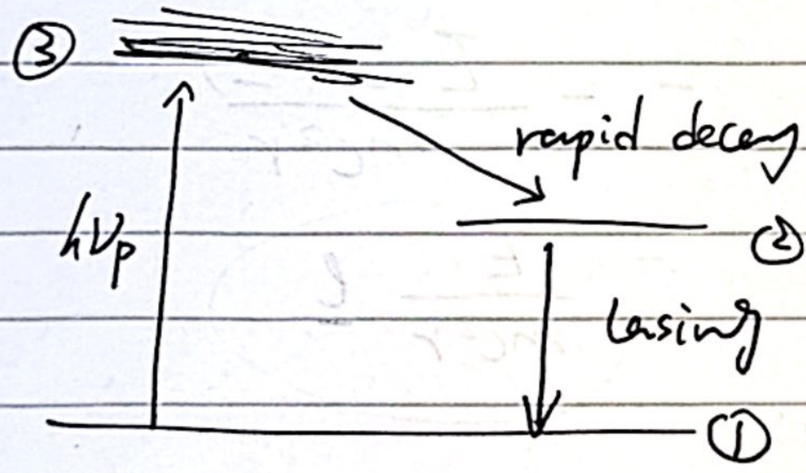
$$E_{th} \approx N_2^{th} V \frac{hc}{\lambda_p} \quad (\sim mJ)$$

(supply $\sim \times 60$ needed practically)

Reasons:
- Only 50% absorbed for uniform pumping
- $\sim$ 50% electrical energy converted to light.
- $\sim$ 10% output of lamp in pump bands.

...

Three-level laser (Ruby)



harder for pop. inv.

$$R_1 R_2 \exp(\sigma N_{th}^* 2L) = 1$$

$N_T = N_2 + N_1$

$N^* = N_2 - N_1$

$$N_2^{th} = \frac{N_T + N_{th}^*}{2} \simeq \boxed{\frac{N_T}{2}}$$

$\left(E_{th} \approx N_2^{th} V \frac{hc}{\lambda_p} \text{ too} \right)$ $(\sim J)$

§ Additions

[X]

S-O coupling derivation

use: $\underline{\mu}_s = -\frac{g_s}{\hbar}\mu_B \underline{S}$

$\hat{H}_{SO} = -\underline{\mu}_s \cdot \underline{B}$

And Thomas precession corrects about $g_s \to g_s - 1$

$$\underline{B} = -\frac{\underline{v} \times \underline{E}}{c^2}$$

$$\simeq -\frac{\underline{v} \times \left(E \frac{\underline{r}}{|\underline{r}|}\right)}{c^2}$$

$$= -\frac{E(\underline{p} \times \underline{r})}{mc^2 r}$$

$$= \frac{E}{mc^2 r}\underline{l}$$

<u>Useful quantities</u>

$m_e \alpha c a_0 = \hbar$

Bohr radius

$\frac{1}{2} m_e \alpha^2 c^2 = hcR_\infty$

Rydberg

Rabi Osc Limit $\Omega_R \gg \frac{1}{\tau}$ ($\sin^2$)

Rate Eqn. Limit $\frac{1}{\tau} \gg \Omega_R$ (Einstein)

Selection Rules (Dipole Approx.)

(in L-S coupling)

$\Delta l = \pm 1$

$\Delta S = 0$

$\Delta L = 0, \pm 1$ (not $0 \leftrightarrow 0$)

$\Delta J = 0, \pm 1$ (not $0 \leftrightarrow 0$)

$\Delta m_J = 0, \pm 1$ (not $0 \leftrightarrow 0$ if $\Delta J = 0$)

Auger Effect

$E_L - E_K + E_L = (E_L - E_K) - (0 - E_L)$

(kinetic energy)